

DAKOTA RENNELS



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Oklahoma City, OK 73170



[Website](#) | [Github Repo](#)

PROFESSIONAL SUMMARY

Lead data scientist and research engineer with extensive experience leading AI/ML projects from concept to production deployment. Proficient in end-to-end deployments, with a focus on LLM-based systems, and streaming-focused data engineering. More than 8 years of experience across analytics, data science, AI/ML, automation, and enterprise software delivery. Experience in public speaking on emerging AI technology to technical and non-technical audiences. Curator of reusable Python libraries to accelerate AI/ML tooling development. Key technical advisor on the Governor's AI Task Force, contributing to AI strategy and policy development for Oklahoma.

SKILLS

- Python, R, SQL, C#, Scala
- Spark & Kafka
- Neural Networks, Deep Learning
- LLMs, RLHFs, Agentic Architecture
- Docker, Helm, & Kubernetes
- Azure, GCP, & AWS
- MLOps, LLMOps, Data Frameworks
- Project Management
- Git & Version Control
- Strong Analytical Skills
- Proficient Statistical Knowledge
- Adept Mentor
- Strong Time Management
- Excellent Team Collaborator
- Strong Business Communication

EDUCATION

University of North Texas

Denton, TX • 12/2023

Master of Science: Advanced Data Analytics

- Graduated ranked 1 out of 1391
- Deep Learning & AI Concentration
- Relevant Coursework: AI Systems, Deep Learning, Data Science, & Statistical Analysis
- 4.0 GPA

WORK HISTORY

Lead Data Scientist and Research Engineer - Paycom

Oklahoma City, OK • 03/2024 - Current

- Serve as company-wide AI architect, advising technical and non-technical departments on AI use case discovery, solution design, implementation strategy, and adoption.
- Own AI roadmap planning across enterprise use cases, prioritizing high-value automation opportunities and aligning delivery with architecture, platform evaluation, governance, and business adoption needs.
- Lead cross-functional AI projects with contributors of 2–20 people across business, data, engineering, and platform teams.
- Develop and deploy end-to-end AI/LLM applications, including LLM workflows, custom-trained models, reinforcement learning designs, and human-in-the-loop feedback patterns.
- Improved client experience with personalization and ranking features using an LLM ranker and Elasticsearch recommendations, reducing mismatch selections by over 80%.
- Author, maintain, and enforce internal AI governance standards aligned to ISO 42001; participate in AI Oversight Committee reviews for responsible AI adoption.

Sr. Data Scientist – Team Lead – Oklahoma Office of Management and Enterprise Services

Oklahoma City, OK • 01/2022 - 03/2024

- Appointed as a technical resource to the Governor's AI Task Force, contributing to AI strategy, policy, and responsible adoption.
- Pioneered the first-ever implementation of a Conversational GenAI chatbot in the United States public sector.

University of North Texas

Denton, TX • 07/2022

Bachelor of Science: Data Analytics

- Graduated ranked 1 out of 1197
- President's Honor Roll: Spring 2022, Fall 2021
- Relevant Coursework: Computer Science, Data Science, & Statistical Analysis
- 4.0 GPA

University Of Oklahoma

Norman, OK

Bachelor of Science: Chemical Engineering

- Relevant Coursework: Engineering, Biochemistry, & Computer Science
- Alpha Sigma Phi Founding Member and Marshal
- Silver Wings President

Oklahoma City Community College

Oklahoma City, OK • 07/2016

Associate of Science: Applied Science

- Relevant Coursework: Computer Science, Economics, & Statistics

CERTIFICATIONS

- Google Encoder-Decoder Architecture
- Google Generative AI for Developers
- AWS Certified Developer
- AWS Cloud Practitioner Certification
- CAP-X Certified
- IBM/Darden Title Specialization
- Six Sigma: Black Belt
- 30+ Data Science and Data Analysis Certifications

- Spearheaded AI & data-driven initiatives resulting in \$2.2 million in cost savings for the state between 2022 and 2024.
- Led a team of 6 data analysts, data scientists, and data engineers designing end-to-end AI workflows.
- Developed AI OCR tool for the Department of Transportation to scan road data for damaged road and property indicators.
- Built and curated Oklahoma land deed parsing tool for historical land deed documents of varying quality.
- Applied A/B testing and quasi-experimental methods to evaluate and improve AI/data science application performance.

Sr. Data Analyst - Sales Operations - Vail Resorts

Broomfield, CO • 10/2020 - 01/2022

- Implemented Python and SQL processes and models for sales-based analytics.
- Performed quality assurance and risk management analysis of internal and external sales systems.
- Enhanced communication methods from the team and other departments.
- Developed improved competitive analysis methodology and coordinated with marketing teams for new method training.
- Lead UAT testing for various e-commerce webpages and internal booking systems.
- Headed process modernization efforts through the implementation of CRISP-DM and DAMA-DMBOK II methodologies for transactional data reports.
- Identified process bottlenecks and inconsistencies to management, facilitating continuous process improvement.

Business Manager - Software - Spirit Shop

Norman, OK • 10/2015 - 06/2020

- Developed and implemented small business automations with Python scripts for enhanced inventory management.
- Analyzed monthly data for sales and inventory trends.
- Monitored day-to-day operations while actively managing project execution.
- Performed daily accounting, budgeting, and transactional analytics.
- Lead teams to provide optimal customer service and lead sales-based events.